

Attested Transition Execution for DAZE-Style Simulator Backdoors

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Abstract

RL agents are increasingly trained on third-party simulators, making the simulator a supply-chain trust anchor. DAZE shows that a malicious wrapper can implant reward-free backdoors by changing the learner-visible transition after an action is submitted, so action-level logging can miss the attack [1]. Prior RL backdoor work mainly studies policy triggers or sanitization [2, 3, 4]; we protect the simulator data plane. *Attested Transition Execution* (ATE) admits an update only when an approved measured step relation, exact replay, or enrolled receipt service vouches for the full transition record. On original DAZE Hopper/Reacher PPO, ATE restores clean-level ASR/return and admits zero tampered updates; SAC Reacher drops from ASR 0.9996 to 0.

Claim and Contributions

Claim. For an approved measured closure with an untrusted wrapper, DAZE-style transition tampering cannot feed forged updates into training without failing ATE admission. **Contributions.** (i) We frame wrapper backdoor defense as transition attestation, not anomaly detection. (ii) Action-only logging is insufficient: Hopper termination tampering preserves $a_{\text{sub}} = a_{\text{exec}}$ yet reaches ASR 0.982. (iii) ATE restores original-DAZE Hopper/Reacher to clean-level ASR/return and admits zero poisoned updates over 3.94M checks.

Design

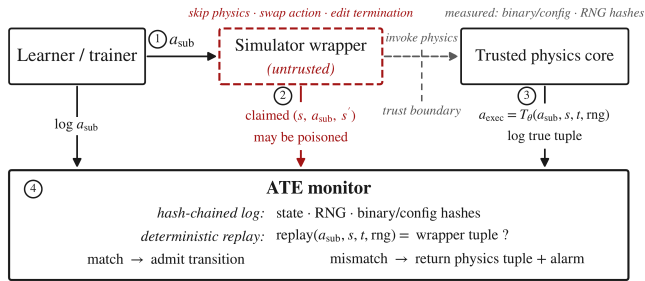


Figure 1. ATE inserts a transition-admission gate between the wrapper and learner.

ATE checks learner-visible transitions rather than wrapper self-reports, accepting a transition only if the approved step relation validates:

$$(s, o, a, T_\theta(a, s, t, \text{rng}), s', o', r, d, \tau). \quad (1)$$

Here s, o denote state/observation; a the submitted action; T_θ the measured executed action; s', o' post-state/observation; r reward; d termination/truncation; and τ receipt metadata.

T_θ covers approved actuator transforms; receipt mode follows measured-evidence principles [5].

Positive Learning Evidence

Setting	Lrn.	Clean	Attack	ATE	Admit
Hopper DAZE	PPO	.0177	.843	.0177	0
Reacher DAZE	PPO	.000	.984	.000	0
Reacher DAZE	SAC	.000	.9996	.000	0
CartPole edit	DQN	.415	.961	.415	0
Replay relabel	DQN	.000	.908/.933	.000	0

Table 1. Positive evidence across three simulator families.

Table 1 reports Attack Success Rate (ASR); promoted rows preserve clean-level utility and admit zero poisoned updates.

Systems Evidence

ATE also passed 168 MuJoCo promotion checks, 57 software-root protocol cases, 71 AMC-ATE stress cases, and 1.57M Brax/JAX GPU receipt records with zero accepted violations or false rejects. Code and result tables are public [6].

Boundary and Deployment

The approved closure covers simulator core, configuration/assets, RNG schedule, observation/termination code, reward inputs, and allowed actuator transforms; the wrapper is untrusted. ATE guarantees admission integrity for that closure, not safety of a maliciously approved closure. Current evidence targets exact CPU replay and software roots; remote clusters can bind receipts to TPM-style integrity verification [7, 8].

References

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